

Alfido Tech — Sales Performance Analysis

Superstore Sales & Order Data · 2015–2018 · Regional, Category, and Seasonal Deep-Dive

Executive Summary

Over the four-year period covered by the dataset (Jan 2015 – Dec 2018), Alfido Tech generated **\$2,261,536.78** in total revenue across **4,922 orders** (9,800 order line items), for an average order value of **\$459.48**. **Technology** is the top revenue category (\$827K) despite having the fewest order lines, driven by high-ticket items such as copiers and networking equipment. The **West** and **East** regions together generate roughly 61% of total revenue, while the **South** region lags all others in both revenue and order volume. Sales are highly seasonal, peaking sharply in **November** and bottoming out in **February** — a swing of nearly 6x between the strongest and weakest month. The source data contains **Sales** (revenue) only; it does not include Profit, Discount, or Quantity fields, so profit margin and unit-based conversion metrics could not be computed directly — this is flagged as a data-collection recommendation below.

Total Revenue	Distinct Orders	Avg Order Value	Avg Ship Lead Time
\$2,261,537	4,922	\$459.48	4.0 days

Note: Profit Margin and Conversion Rate are not shown above — the source data has no Profit, Discount, Quantity, or traffic/lead columns to derive them from (see Section 5).

1. Data Overview & Cleaning

The dataset contains 9,800 order line items across 18 original columns (Order/Ship dates, Ship Mode, Customer, Segment, Region/State/City, Product hierarchy, and Sales). Data quality was high: **zero duplicate rows** and only **11 missing values** (0.1%), confined to Postal_Code, which is not required for the regional analysis below. Dates were parsed as DD/MM/YYYY (confirmed by day-of-month values exceeding 12), and derived fields (Order Year/Month, Year-Month period, Shipping Lead Time) were added to support trend and seasonality analysis.

Metric	Sales (\$)	Ship Lead Time (days)
Mean	230.77	3.96
Median	54.49	4.00
Mode	12.96	4.00
Std Dev	626.65	1.75
Min / Max	0.44 / 22,638.48	0 / 7

Table 1: Summary statistics for numeric fields (full detail — including per-column data types and mode/frequency for categorical fields such as Region, Category, and Ship Mode — is in the companion notebook, Section 2 & 4).

2. Core Visualizations

2.1 Revenue by Category

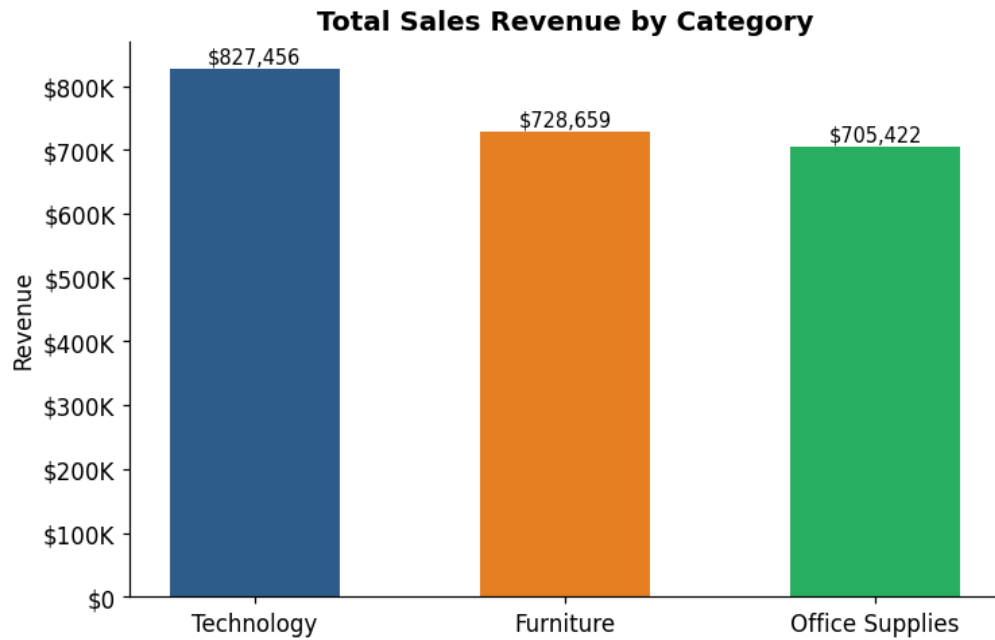


Figure 1: Technology leads on total revenue (\$827K), narrowly ahead of Furniture (\$729K) and Office Supplies (\$705K), despite Office Supplies having by far the most order lines.

2.2 Monthly Revenue Trend

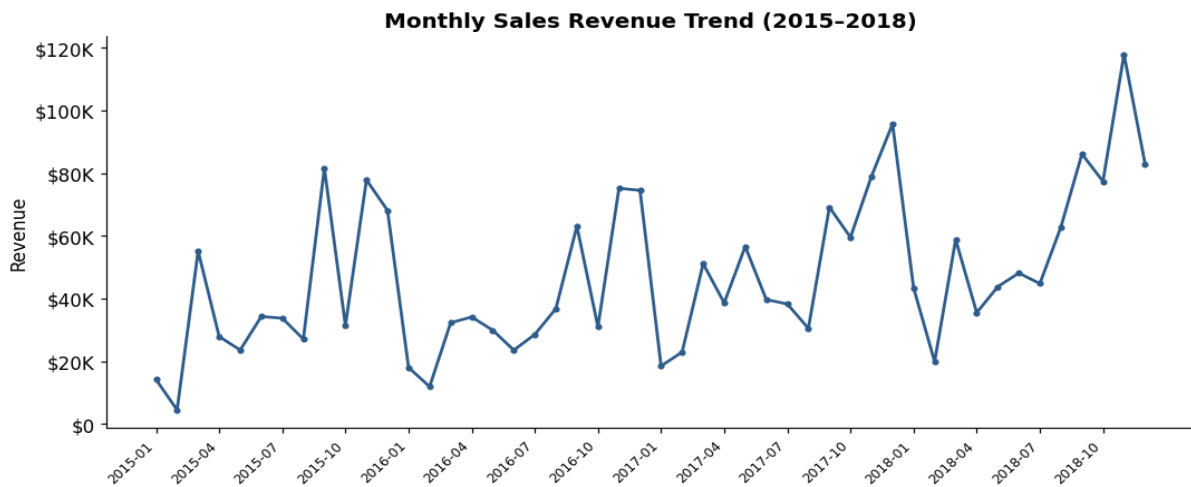


Figure 2: Revenue trends upward year over year with a recurring seasonal spike each Sep-Dec, most pronounced in the final year.

2.3 Distribution of Order-Line Sales Value

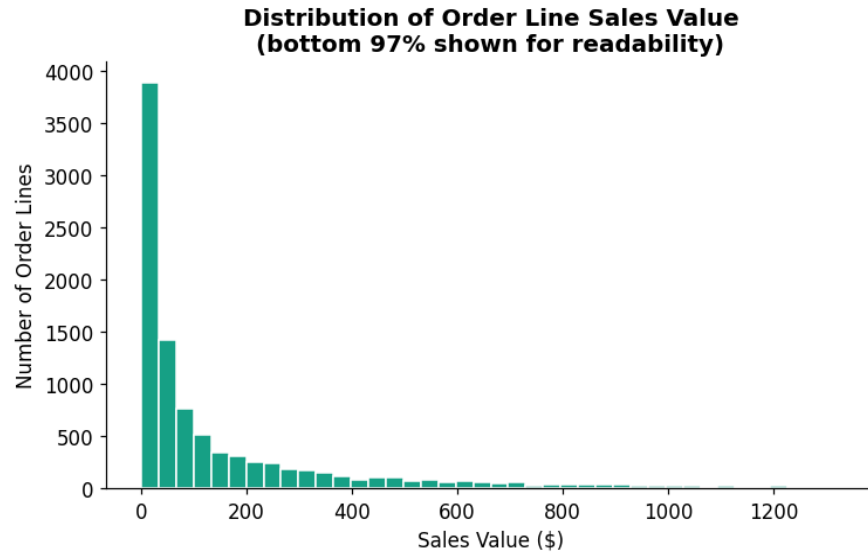


Figure 3: Order values are heavily right-skewed — most line items are small-ticket (under ~\$100), with a long tail of high-value Technology and Furniture purchases (top 3% trimmed for readability).

3. Regional & Category Performance

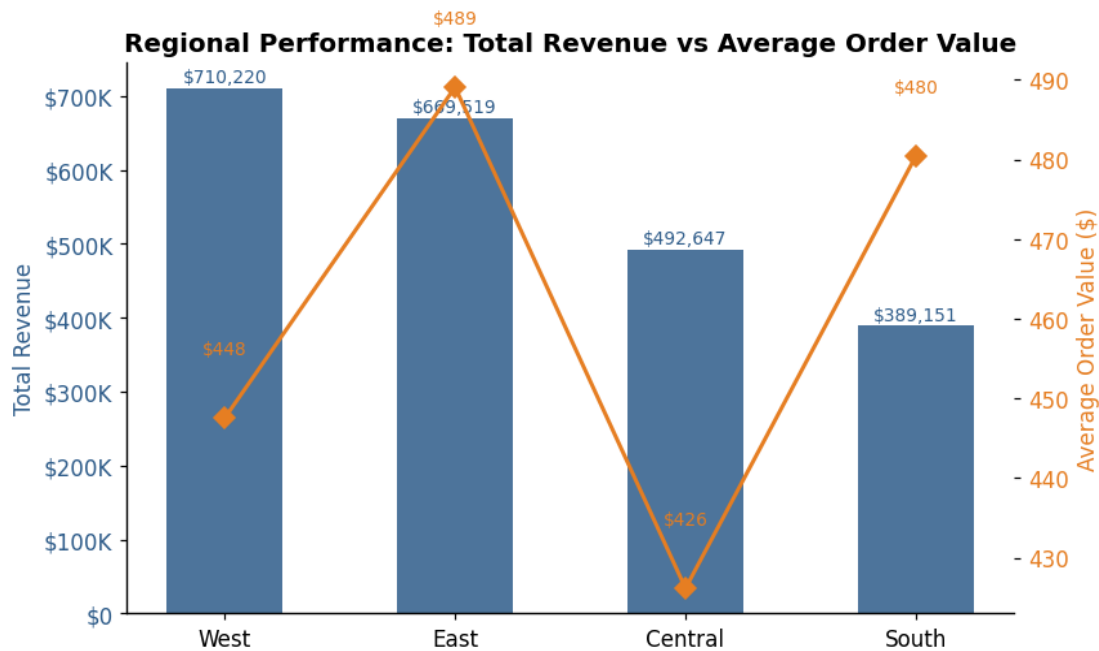


Figure 4: The West (\$710K) and East (\$670K) regions lead on total revenue, while the South (\$389K) lags all regions on volume even though the East has the highest average order value (\$489).

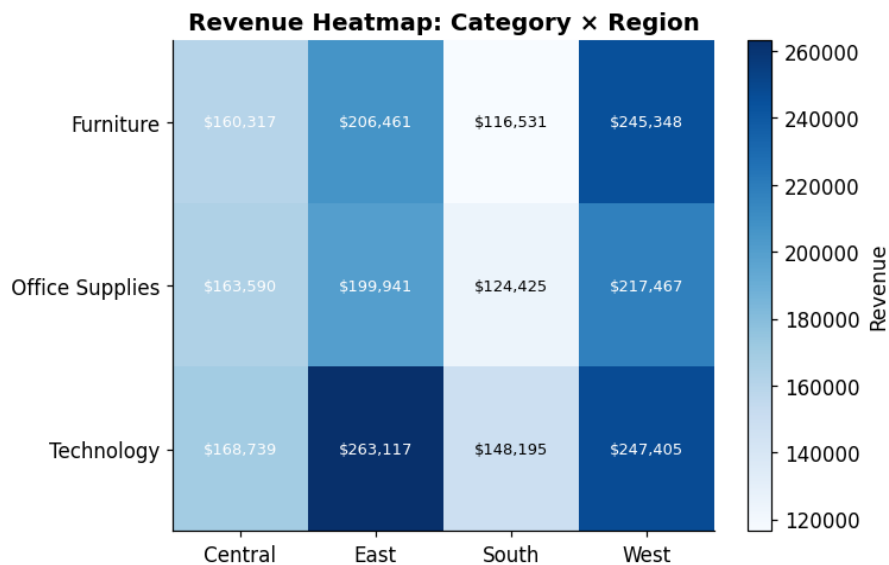


Figure 5: Category × Region revenue heatmap — the South region under-indexes across all three categories, most notably in Technology and Furniture.

Revenue Share by Customer Segment

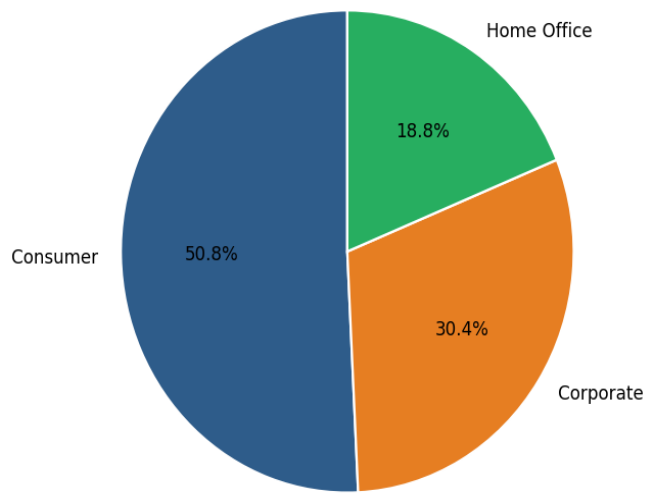


Figure 6: The Consumer segment drives roughly half of total revenue, ahead of Corporate and Home Office.

4. Best/Worst Products & Seasonality

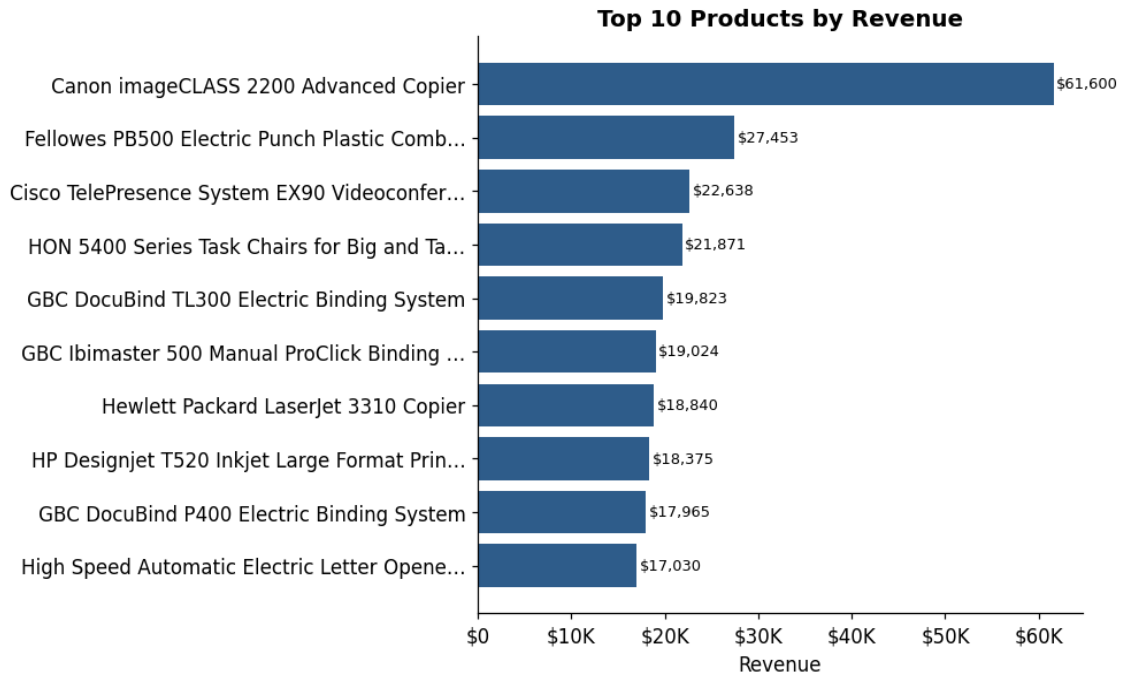


Figure 7: The Canon imageCLASS 2200 Copier alone generated \$61.6K — more than double the next-highest product. The top 10 products are dominated by big-ticket Technology and binding/office machines.

Lowest-revenue products (bottom of the tail) are small consumables such as paper clips, highlighters, and letter openers — individually generating under \$10 in total revenue across the full 4-year period, and are candidates for catalog rationalization or bundling rather than standalone promotion.

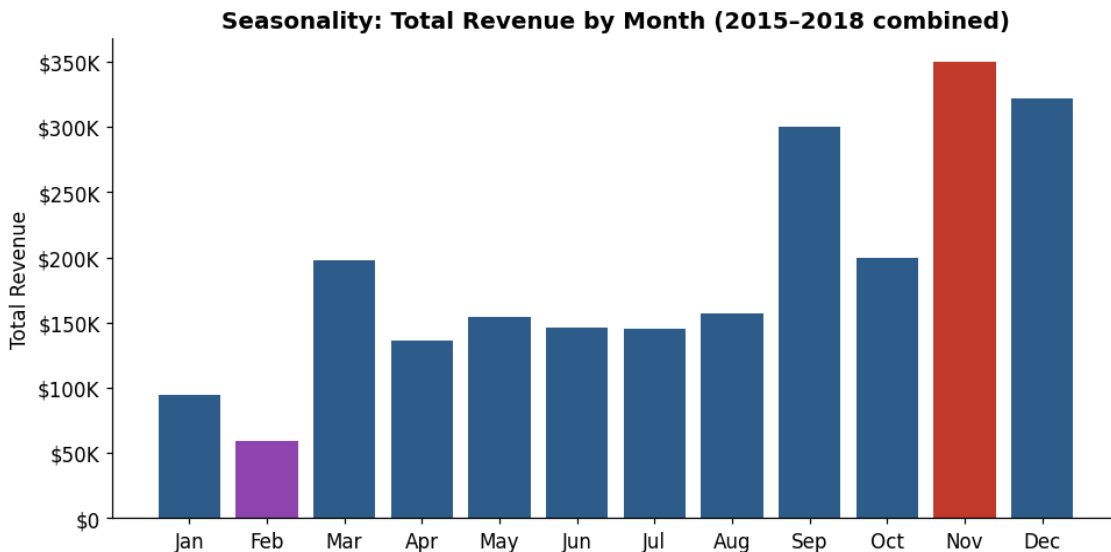


Figure 8: Seasonality — revenue climbs steadily from spring, surges Sep–Dec (peaking in November at \$350K combined across all years), and troughs in February (\$59K), a ~6x swing.

5. KPI Methodology & Data Gaps

KPI	Value / Status	Method
Total Revenue	\$2,261,536.78	Sum of Sales
Average Order Value (AOV)	\$459.48	Total Revenue ÷ Distinct Orders (Order_ID)
Average Line-Item Value	\$230.77	Mean of Sales per row
Avg Shipping Lead Time	4.0 days	Ship_Date – Order_Date
Profit Margin	Not available	No Profit or Cost column in source data
Conversion Rate	Not available	No site-visit/lead/quote-stage data in source

Recommendation: extend the sales pipeline to capture Quantity, Discount, and Cost per unit so true Profit Margin can be tracked, and connect marketing/website analytics so a genuine funnel Conversion Rate can be measured against these orders.

6. Three Actionable Insights

- 1 **Double down on Technology in underperforming regions.** Technology drives the highest revenue per line item, but the South region under-indexes on Technology sales relative to West/East. A targeted push — bundling, promotions, rep incentives — on Technology in the South could close a meaningful share of the regional revenue gap.
- 2 **Plan inventory and marketing around the Sep–Nov surge.** With November revenue roughly 6x February's, Alfido Tech should pre-build stock and launch promotional campaigns starting August/September to be ready for the demand ramp, and use the Jan–Feb trough for clearance pricing on slow-moving stock.
- 3 **Protect and grow the low-frequency, high-value product tail.** The top 10 products (copiers, binding machines, big-ticket furniture) generate a disproportionate share of revenue relative to their order count. These deserve premium placement, dedicated upsell/cross-sell paths, and stock-out prevention.

7. Five Tactical Improvements

- 1 **Regional rebalancing:** launch a dedicated South-region campaign (local promotions, incentives, or a regional distribution partner) to close the ~45% revenue gap versus the West.
- 2 **Seasonal inventory & staffing plan:** build a demand-driven restocking calendar with safety-stock increases starting August ahead of the Sep–Nov peak, and discount excess stock in Jan–Feb.
- 3 **Bundle high-margin Technology with high-frequency Office Supplies** (e.g. printer + paper + toner) to lift average order value across both categories.
- 4 **Introduce a loyalty/repeat-purchase program for the Consumer segment** (the largest revenue segment) to convert one-time buyers into repeat customers.

- 5 **Instrument true conversion & profitability tracking:** add Quantity, Discount, and Cost/Profit fields to the sales pipeline — without them, true margin and funnel conversion cannot be measured, limiting how precisely marketing spend and discounting can be optimized.

Source: `superstore_final_dataset.csv` (9,800 order line items, Jan 2015–Dec 2018). Full code, cleaning steps, and additional charts are available in the companion Jupyter notebook *Alfido_Tech_Sales_Analysis.ipynb*.